

Unified Military Analytics

Project Overview:

Unified Military Analytics is an interactive military data visualization project that provides a unified view of military strength, defense capabilities, economic indicators, demographics, and geopolitical information for countries around the world.

The project combines data collection, data cleaning, KPI feature engineering, and interactive dashboard development to support country-level analysis and comparison.

The final solution contains four interactive dashboards:

1. Quick Stats
2. Nation Overview
3. Compare Powers
4. Defense and Navy

Project Objectives:

The main objectives of this project are:

- Collect military data from Global Firepower.
- Clean and standardize military datasets.
- Combine military data with demographic and economic information.
- Add region, continent, and alliance information.
- Engineer useful military comparison KPIs.
- Build interactive dashboards.
- Provide country-level military analysis.
- Enable comparison between two countries.
- Create a clean, documented, and shareable analytics project.

Data Sources:

The primary military data source used in the project is **Global Firepower**.

The dataset contains country-level military, demographic, economic, and capability-related indicators.

Major Military Indicators:

- Power Index
- Power Index Rank
- Active Personnel
- Reserve Personnel
- Paramilitary Personnel
- Total Military Aircraft
- Fighter Aircraft
- Attack Aircraft
- Transport Aircraft
- Trainer Aircraft
- Special Mission Aircraft
- Tanker Aircraft
- Military Helicopters
- Attack Helicopters
- Tanks
- Armored Fighting Vehicles
- Self-Propelled Artillery
- Towed Artillery
- Rocket Projectors
- Total Naval Fleet
- Naval Fleet Tonnage
- Aircraft Carriers
- Helicopter Carriers
- Submarines
- Destroyers
- Frigates
- Corvettes
- Coastal Patrol Craft
- Mine Warfare Craft

- Defense Budget

Data Collection:

Military data was collected using a Python-based scraping workflow.

Technologies Used:

- Python
- BeautifulSoup
- Pandas
- Excel/CSV

The scraping process collected the required country-level military indicators from the relevant Global Firepower pages.

The raw information was then combined into the project's military dataset.

Data Cleaning:

The raw data was processed before dashboard development.

The cleaning workflow included:

- Standardizing column names.
- Standardizing country names.
- Removing commas from numeric values.
- Removing unnecessary special characters.
- Removing percentage symbols where required.
- Converting numeric fields into appropriate numeric data types.
- Handling missing and null values.
- Combining datasets using country as the common field.
- Adding continent and region information.
- Adding alliance information.
- Validating the final dataset.
- Preparing the data for dashboard visualization.

KPI Feature Engineering:

Several KPIs were created to support military analysis and comparison.

Power Index Rank Gap

Represents the difference between the military rankings of two countries.

Assets per Capita

Measures military assets relative to the country's population.

Formula:

$$\text{Assets per Capita} = \text{Total Military Assets} / \text{Population}$$

Budget-to-GDP Ratio

Measures defense spending as a percentage of GDP.

Formula:

$$\text{Budget-to-GDP Ratio} = (\text{Defense Budget} / \text{GDP}) \times 100$$

Power Advantage:

Represents the relative military power difference between two selected countries.

Budget Advantage

Represents the relative defense-budget difference between two selected countries.

Combined Personnel

Represents the combined personnel strength of selected countries.

Coalition Budget

Represents the combined defense budget of selected countries.

Coalition Score

Represents a combined measure of military capability for coalition analysis.

Dashboard Overview:

Dashboard 1 — Quick Stats

Quick Stats provides a high-level overview of the world's leading military powers.

Main Features:

- Top 10 military powers.
- Power Index information.
- Dynamic KPI cards.
- Region filter.
- Continent filter.
- Alliance filter.

- Interactive charts.

This dashboard is intended as the starting point for exploring the dataset.

Dashboard 2 — Nation Overview

Nation Overview provides a detailed profile of a selected country.

Main Information:

- Country
- GDP
- Defense Budget
- Active Personnel
- Military Assets
- Air capabilities
- Naval capabilities
- Power Index
- Power Index Rank

Users can select a country and view its military and economic profile through KPI cards and charts.

Dashboard 3 — Compare Powers

Compare Powers allows users to compare two countries.

Users select:

- Country A
- Country B

The dashboard displays comparative military and economic indicators.

Dashboard 4 — Coalition Builder

Defense and Navy focuses on military asset categories related to defense and naval capabilities.

The dashboard allows users to inspect military assets and naval measures through interactive visualizations and tooltips.

Navigation of Dashboards:

The recommended navigation flow is:

Quick Stats → Nation Overview → Compare Powers → Coalition Builder

Navigation controls are provided within the workbook to move between dashboards.

Testing the Dashboard:

The completed workbook was tested during Module 7.

Testing covered:

- Dashboard functionality.
- Filters.
- Parameters where applicable.
- Dashboard navigation.
- KPI calculations.
- Data accuracy.
- Tooltips.
- Axis visibility.
- Labels.
- Dashboard layout.
- Workbook loading.

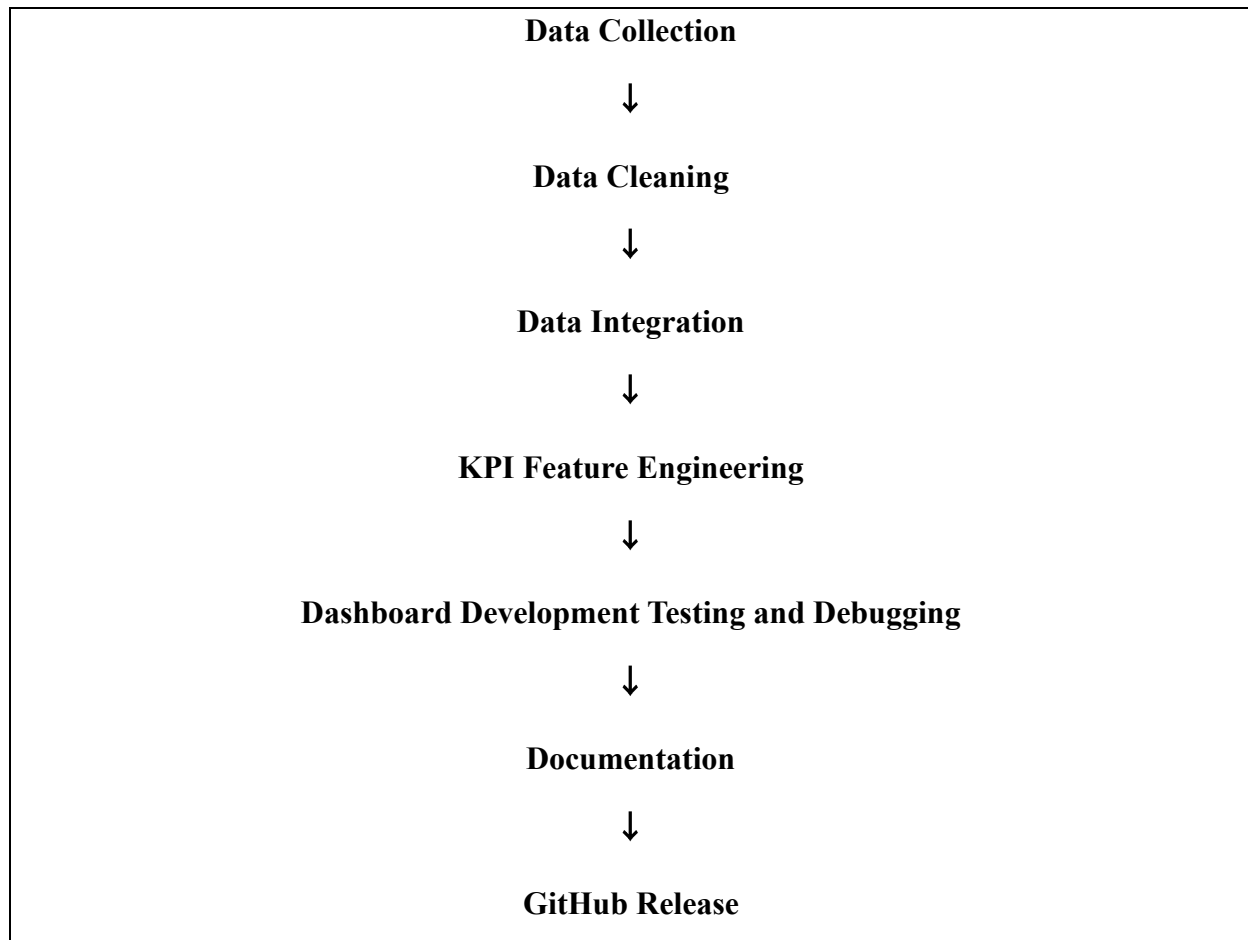
All four dashboards were reviewed and confirmed to be functioning correctly.

Technologies:

- Python
- Pandas
- BeautifulSoup
- Microsoft Excel
- Tableau Desktop
- GitHub

Project Workflow:

The overall project workflow was:



Documentation:

The repository contains the following documentation:

README

Provides an overview of the project, data, KPIs, dashboards, technologies, and usage.

QA Checklist

Contains the Module 7 testing and validation checklist